Preliminaries

- We consider the following set of equations

$$\boldsymbol{Ax} = \boldsymbol{b}$$

$$\boldsymbol{A}(m \times n);\ \boldsymbol{x}(n \times 1);\ \boldsymbol{b}(m \times 1)$$

- Generalized linear equations can be represented in the above format.
- m and n are the number of equations and variables respectively.
- $\boldsymbol{b}$ is the general RHS commonly used

Categorization

$m = n$

- Number of equations and variables are the same
- Easiest case to solve

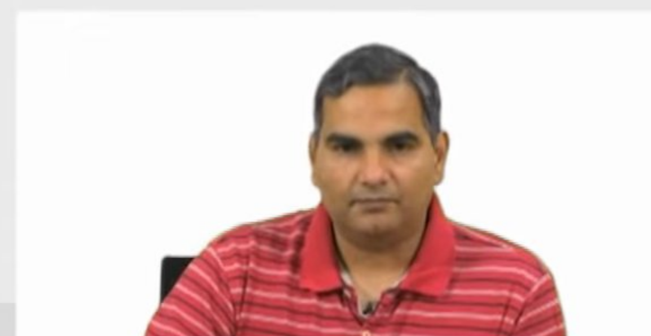
$m > n$

- More equations than variables
- Usually no solution

$m < n$

- Number of equations less than number of variables
- Usually multiple solutions

We look into these cases independently



Full row and column rank: Concepts

- Consider a matrix data matrix $\boldsymbol{A}$ $(m \times n)$

Full Row Rank

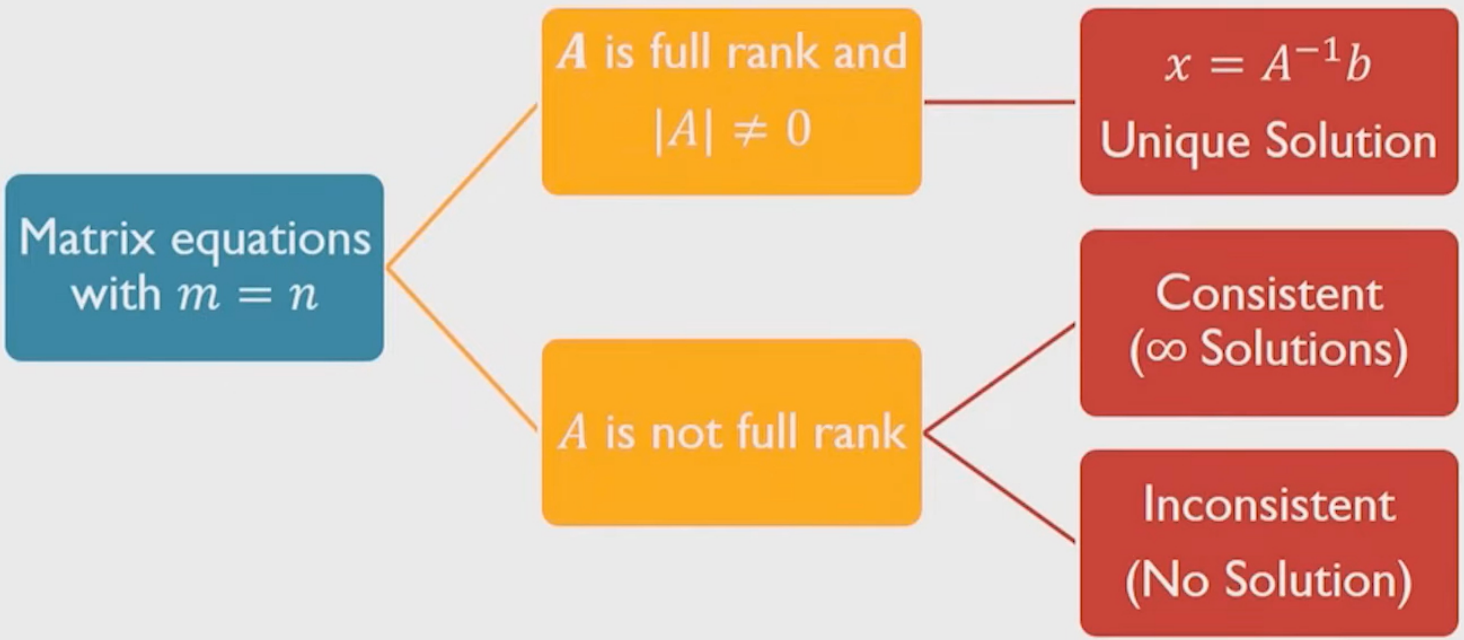
- When all the rows of the matrix are linearly independent
- Data sampling does not present a linear relationship – samples are independent

Full Column Rank

- When all the columns of the matrix are linearly independent
- Attributes are linearly independent

Row rank = Column rank

Case 1: $m = n$


Matrix equations with $m = n$
A is full rank and $|A| \neq 0$
$x = A^{-1}b$
Unique Solution
A is not full rank
Consistent (∞ Solutions)
Inconsistent (No Solution)

Case 1: Example 1.1

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}$$

R Code

```
A=matrix(c(1,2,3,4),ncol=2, byrow=F)
b=c(7,10)
x=solve(A)%*%b
```

$$|\boldsymbol{A}| \neq 0$$

$$rank(\boldsymbol{A}) = 2 = no.of\ columns$$

Console output

```
> x
     [,1]
[1,]    1
[2,]    2
```

- This implies that A is full rank

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 7 \\ 10 \end{bmatrix} = \begin{bmatrix} -2 & 1.5 \\ 1 & -0.5 \end{bmatrix} \begin{bmatrix} 7 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (1,2)$

Case 1: Example 1.2

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$|\boldsymbol{A}| = 0;\ rank(\boldsymbol{A}) = 1; nullity = 1$

- Checking consistency

$$\begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$Row(2) = 2Row(1)$$

- The equations are consistent with only one linearly independent equation
- The solution set for $(x_1, x_2\ )$ is infinite because we have only one linearly independent equation and 2 variables

Case 1: Example 1.3

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

$$|\boldsymbol{A}| = 0$$
$$rank(\boldsymbol{A}) = 1$$
$$nullity = 1$$

- Checking consistency

$$\begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$
$$2Row(1) = 2x_1 + 4x_2 = 10 \neq 9$$

- Thus the equations are inconsistent
- One cannot find a solution to (x_1, x_2)

Case 2: $m > n$

- This is the case of not enough variables or attributes
- Since the number of equations is greater than the number of variables, in general, not all equations can be satisfied
- Hence it is sometimes termed as a no-solution case
- However, we can identify an appropriate solution by viewing this case from an optimization perspective

Case 2: An optimization perspective

- Instead of identifying a solution to $Ax - b = 0$, one can identify an x such that $(Ax - b)$ is minimized
- Notice that $(Ax - b)$ is a vector
- There will be as many error terms as the number of equations
- Denote $(Ax - b) = e(mx1)$; there are m errors $e_i, i = 1{:}\, m$
- One could minimize all the errors collectively by minimizing $\sum_{i=1}^{m} e_i^2$
- This is the same as minimizing $(Ax - b)^T(Ax - b)$

Case 2: An optimization perspective

- This optimization problem is

$$\min[(Ax-b)^T(Ax-b)]$$
$$=\min[(b^T-x^TA^T)\,(Ax-b)]$$
$$=min[(x^TA^TAx-2b^TAx+b^Tb)=f(x)]$$

- We observe that the optimization problem is a function of x
- Solving the optimization problem will result in a solution for x
- The solution to this optimization problem is obtained by differentiating $f(x)$ with respect to x and setting the differential to zero

$$\nabla f(x)=0$$

Case 2: An optimization perspective

- Differentiating $f(x)$ and setting the differential to zero results in

$$2(A^T A)x - 2A^T b = 0$$
$$(A^T A)x = A^T b$$

- Assuming that all the columns are linearly independent

$$\boldsymbol{x} = (\boldsymbol{A}^T \boldsymbol{A})^{-1} \boldsymbol{A}^T \boldsymbol{b}$$

Case 2: Example - I

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

- $m = 3, n = 2$
- Using the optimization concept,

$$\boldsymbol{x} = (\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T\boldsymbol{b}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

Case 2: Example continued

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.6 \\ -0.6 & 2.8 \end{bmatrix} \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (0,5)$
- Substituting in the equation shows

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \neq \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

Case 2: Example

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

- $m = 3, n = 2$
- Using the optimization concept,

$$\boldsymbol{x} = (\boldsymbol{A}^T \boldsymbol{A})^{-1} \boldsymbol{A}^T \boldsymbol{b}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_2 \end{bmatrix} \quad \begin{bmatrix} -0.6 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

- Thus, the solution for t
$(x_1, x_2) = (1,2)$

Case 2: Example continued

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.6 \\ -0.6 & 2.8 \end{bmatrix} \begin{bmatrix} 20 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (1,2)$
- Substituting in the equation shows

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

R Code

```
A=matrix(c(1,2,3,0,0,1),ncol=2, byrow=F)
b=matrix(c(1,2,5),ncol=2, byrow=F)
x=inv(t(A)%*%A)%*%t(A)%*%b
x
```

Console output

```
> x=inv(t(A)%*%A)%*%t(A)%*%b
> x
     [,1]
[1,]    1
[2,]    2
>
```

Case 3: $m < n$

- This case addresses the problem of more attributes or variables than equations
- Since the number of attributes is greater than the number of equations, one can obtain multiple solutions for the attributes
- This is termed as an infinite-solution case
- How does one choose a single solution from the set of infinite possible solutions?

Case 3: An optimization perspective

- Pose the following optimization problem

$$\min\left(\frac{1}{2}x^T x\right) \text{ s.t. } Ax = b$$

- Define a Lagrangian function $f(x, \lambda)$

$$\min\left[f(x, \lambda) = \frac{1}{2}x^T x + \lambda^T (Ax - b)\right]$$

- Differentiating the Lagrangian with respect to x, and setting to zero

$$x + A^T \lambda = 0$$

Case 3: An optimization perspective

$$x = -A^T\lambda$$

Pre-multiplying by A

$$Ax = b = -AA^T\lambda$$

Thus we obtain $\lambda = -(AA^T)^{-1}b$ assuming that all the rows are linearly independent

$$\boldsymbol{x} = -\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{\lambda} = \boldsymbol{A}^{\boldsymbol{T}}\left(\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}}\right)^{-1}\boldsymbol{b}$$

Case 3: Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- $m = 2, n = 3$
- Using the optimization concept,

$$\boldsymbol{x} = \boldsymbol{A}^{\boldsymbol{T}} \left(\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}} \right)^{-\mathbf{1}} \boldsymbol{b}$$

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Case 3: Example

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 14 & 3 \\ 3 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -0.2 \\ 1.6 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -0.2 \\ -0.4 \\ 1 \end{bmatrix}$$

- The solution for the given example is $(x_1, x_2, x_3) = (-0.2, -0.4, 1)$

R Code

```
A=matrix(c(1,0,2,0,3,1),ncol=3)
b=c(2,1)
library(MASS)
x = t(A)%*%inv(A%*%t(A)) %*%b
x
```

Console output

```
A=matrix(c(1,0,2,0,3,1),ncol=3, byrow=F)
b=c(2,1)
x = t(A)%*%inv(A%*%t(A)) %*%b
x
```

Case 3: Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- The solution for the given example is $(x_1, x_2, x_3) = (-0.2, -0.4, 1)$
- Verify this is a solution that satisfies the original equation
- This also turns out to be minimum norm solution

Generalization

- The described cases cover all the scenarios one might encounter while solving linear equations
- Is there any form in which the results obtained for cases 1, 2 and 3 can be generalized ?
- The concept we used to generalize the solutions is called as Moore-Penrose pseudo-inverse of a matrix
- The pseudo inverse is used as follows

$$Ax = b$$

The solution becomes

$$x = A^{+}b$$

- Singular Value Decomposition can be used to calculate the pseudo inverse or the generalized inverse (A^{+})

Two examples revisited

Example 2

R Code

```
A=matrix(c(1,2,3,0,0,1),ncol=2, byrow=F)
b=matrix(c(1,2,5),ncol=1, byrow=F)
library(MASS)
x= ginv(A)%*%b
```

Solution

```
> x [,1]
[1,] 1
[2,] 2
```

Example 3

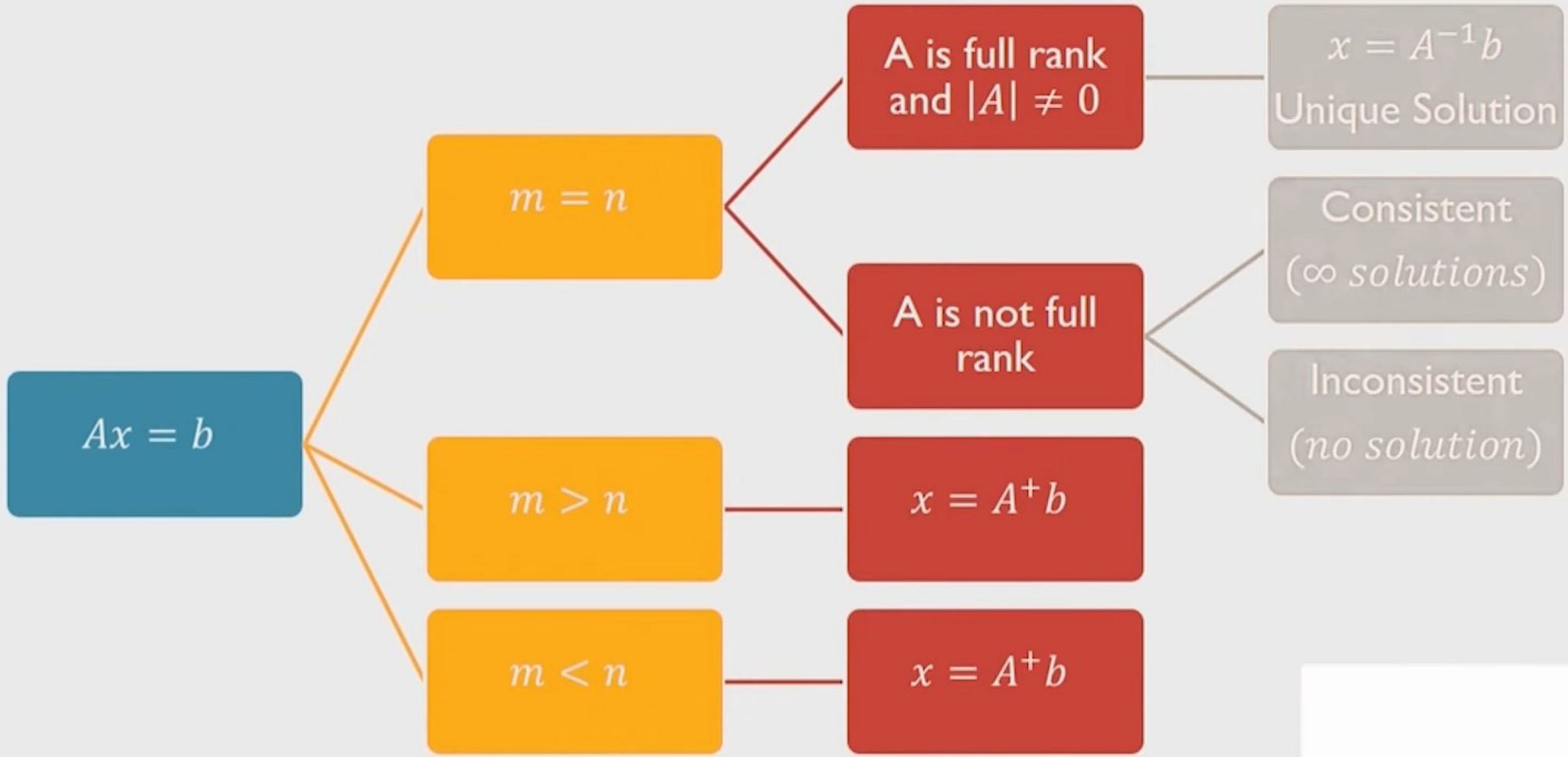
R Code

```
A=matrix(c(1,0,2,0,3,1),ncol=3, byrow=F)
b=c(2,1)
library(MASS)
x = ginv(A)%*%b
```

Solution

```
> x
     [,1]
[1,] -0.2
[2,] -0.4
[3,]  1.0
```

Summary



where we define pseudo inverse A^{+} appropriately